

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 20%

QUESTIONS: 05

Total Marks:50

Annexure A

COMPARITIVE ANALYSIS

Criteria	Scope of Items (2)	Comparison Depth(2.5)	Use of Evidence (2)	Critical Thinking (2)	Clarity of Presentation (1)	Total(10)
<i>Weightage</i>	20%	25%	20%	20%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I M MEGHANADH(192325026) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MEGHANADH

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Q1 . Dataset: Assessment Tool 2: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes
Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data

Customer Income (₹) Spending Score

C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:

Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. *(BL3 – 1 Mark)*

Q2. Dataset: Document Data

Document Word1 Word2 Word3

D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:

Compare K-means and EM algorithms in handling overlapping clusters for the given dataset. *(BL4 – 1 Mark)*

Q3. Dataset: Geographic Data

Point Latitude Longitude

P1	13.08	80.27
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P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

Question:

Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. *(BL4 – 1 Mark)*

Q4. Dataset: Retail Products

Product Sales (₹) Frequency

P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question:

Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. *(BL3 – 1 Mark)*

Q5 . Dataset: Mixed Attributes

Item Price (₹) Category Rating

I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question:

Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. *(BL4 – 1 Mark)*

Q1. K-Means vs Hierarchical Clustering

The dataset contains **Customer Income** and **Spending Score**. Both K-Means and Hierarchical Clustering can group customers with similar characteristics.

Aspect	K-Means	Hierarchical Clustering
Method	Divides data into a predefined K number of clusters	Builds a hierarchy/dendrogram of clusters
Cluster selection	K must be chosen beforehand	Number of clusters can be selected by cutting the dendrogram
Performance	Generally faster for larger datasets	Usually slower as dataset size increases
Scalability	Highly scalable and suitable for large customer datasets	Less scalable for very large datasets
Interpretation	Easy to understand cluster centers	Dendrogram shows relationships between customers

For this dataset, K-Means can quickly group customers such as **C1 and C4** (lower income with higher spending) and **C2 and C5** (higher income with lower spending). Hierarchical clustering can show these relationships through a dendrogram.

Conclusion: K-Means is generally better for **large-scale customer segmentation because of its speed and scalability**, while hierarchical clustering is useful when the company wants to **visualize the relationships between customers and explore different numbers of clusters**.

Q2. K-Means vs EM for Overlapping Clusters

For the document dataset, **K-Means** assigns each document to only one cluster, so it is less suitable when documents have overlapping word distributions.

EM (Expectation-Maximization) assigns **probabilities of membership** to different clusters. Therefore, it can better handle documents that share similar word patterns.

For example, **D1, D2, and D5** have similar Word2/Word3 distributions, while **D3 and D4** are strongly dominated by Word1.

Conclusion: **EM is more suitable for overlapping document clusters** because it provides probabilistic membership, whereas K-Means makes a hard assignment.

Q3. Euclidean vs Manhattan Distance

Euclidean distance measures the straight-line distance between two geographic points:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Manhattan distance measures distance along the coordinate axes:

$$d = |x_2 - x_1| + |y_2 - y_1|$$

For the given geographic data, **Euclidean distance** is generally more appropriate because latitude and longitude represent spatial coordinates, and straight-line proximity is useful for geographic clustering.

Impact on clustering: Euclidean distance tends to group geographically close points such as **P1, P2, and P4**, while Manhattan distance can produce slightly different cluster boundaries because it adds the absolute differences along each coordinate.

Conclusion: The choice of distance metric can change cluster membership. For geographic point clustering, **Euclidean distance is generally preferred** when straight-line spatial proximity is the desired measure.

Q4. Agglomerative vs Divisive Clustering

- **Agglomerative clustering** starts with each product as a separate cluster and **merges similar products** step by step.
- **Divisive clustering** starts with all products in one cluster and **splits them** into smaller clusters.

For this dataset, products such as **P1 and P3** have similar high sales and frequency, while **P2, P4, and P5** have relatively lower sales and frequency.

Conclusion: Agglomerative clustering is more suitable because the dataset is small and it can easily identify similar product groups through a dendrogram. Divisive clustering is also possible but is generally more computationally expensive.

Q5. Centroid-Based vs Density-Based Clustering

Aspect	Centroid-Based	Density-Based
Method	Groups data around cluster centers	Groups dense regions of data
Example	K-Means	DBSCAN
Mixed data	Needs numerical encoding for Category	Also requires suitable distance handling
Outliers	Can be sensitive to outliers	Can identify outliers as noise
Cluster shape	Works best with roughly spherical clusters	Can detect irregular-shaped clusters

For this dataset, **Price and Rating are numerical**, while **Category is categorical**. K-Means cannot directly process Category, so it must first be appropriately encoded. Density-based methods also require a suitable distance measure capable of handling the mixed numerical and categorical attributes.

The data shows a clear pattern: **I1, I3, and I5** belong to Category A with higher prices and ratings, while **I2 and I4** belong to Category B with lower prices and ratings.

Conclusion: For this small mixed-type dataset, neither standard K-Means nor standard DBSCAN should be applied directly. **Appropriate encoding and a mixed-data distance measure are required.** Centroid-based clustering is suitable when the data is properly transformed, while density-based clustering is useful when irregular clusters and outliers are important.